THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET A

Chapter 2: **Introduction to Linear Polynomials** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The degree of the polynomial $3x^2 - 5x + 7$ is: 1

(a) 3 **(b)** 1 **(c)** 2 **(d)** 4

2. The degree of the polynomial $x^4 - 3x^3 + 6x^2 - 2x + 7$ is: 1

(a) 4 **(b)** 6 **(c)** 5 **(d)** 3

3. The value of the linear polynomial $-4x + 7$ at $x = 3$ is: 1

(a) -5 **(b)** -9 **(c)** -19 **(d)** 5

4. The value of $7x^2 - 4x + 6$ at $x = -3$ is: 1

(a) 83 **(b)** 57 **(c)** 78 **(d)** 81

5. The sequence $5, 7, 9, 11, \ldots$ forms a linear pattern. Its nth term is: 1

(a) $2n - 7$ **(b)** $2n + 3$ **(c)** $2n + 5$ **(d)** $5n + 2$

6. For the line $y = 2x + 5$, the slope and the y-intercept are respectively: 1

(a) 5 and 2 **(b)** -2 and 5
(c) 2 and 5 **(d)** 2 and -5

7. For the line $y = -x - 2$, the slope and the y-intercept are respectively: 1

(a) -1 and 2 **(b)** 1 and -2
(c) -2 and -1 **(d)** -1 and -2

8. The point $(3,\ 1)$ ___ the line $y = -x + 4$. 1

(a) lies on the line **(b)** is the origin
(c) does not lie on the line **(d)** is the y-intercept of the line

9. The constant term of $9x^3 + 5x^2 - 8x - 10$ is: 1

(a) 5 **(b)** 9 **(c)** −10 **(d)** −8

10. The graph of $y = -2x$ rises or falls as x increases? 1

(a) is horizontal **(b)** rises
(c) is vertical **(d)** falls (negative slope)

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A linear pattern begins $4,\ 7,\ 10, \ldots$ Write the expression for its nth term and find the 10th term. 2

12. A linear pattern begins $2,\ 7,\ 12, \ldots$ Write the expression for its nth term and find the 12th term. 2

13. A linear pattern begins $100,\ 95,\ 90, \ldots$ Write the expression for its nth term and find the 8th term. 2

14. A linear pattern begins $7,\ 11,\ 15, \ldots$ Write the expression for its nth term and find the 15th term. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The present age of Salil's mother is three times his age. After 5 years their ages will add up to 70. Find their present ages. 3

16. The difference between two positive integers is 63 and their ratio is 2 : 5. Find the integers. 3

17. Ruby has 3 times as many ₹2 coins as ₹5 coins. If she has ₹88 in all, how many coins of each type does she have? 3

OR

A farmer cuts a 300 ft fence into two pieces; the longer is four times the shorter. Find the lengths. 3

18. The length of a rectangle is 3 more than twice its width and its perimeter is 24 cm. Find its dimensions. 3

OR

A rally starts with 120 members and 9 drop out every hour. Find a linear expression for the number remaining at the end of the nth hour and the number after 7 hours. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A taxi company charges ₹100 as a fixed fee and ₹60 per km. The cost for d km is $C(d) = 100 + 60d$. 5

(i) Find the cost for 5 km and 15 km.

(ii) For how many km is the cost ₹700?

(iii) Is this linear growth or decay? Write the slope and y-intercept and what each means.

OR

(i) Plot the graphs of $y = 2x - 1$, $y = 2x + 1$ and $y = 2x + 5$ on the same axes by choosing points. 5

(ii) What do you observe about the lines?

(iii) Where does $y = 2x + 5$ cut the y-axis?

20. **Case study:** A growing pattern of tiles has 1, 3, 5, 7, … tiles in stages 1, 2, 3, 4. 5

(i) Find the expression for the number of tiles in stage n.

(ii) How many tiles are there in stages 15 and 26?

(iii) Which stage has 21 tiles? Which has 47?

(iv) What is the slope of the line $y = 2x - 1$ and what does it represent?

OR

A telecom company charges a fixed fee and a rate per GB. When 10 GB are used the bill is ₹350; for 20 GB it is ₹550. 5

(i) Find a and b in $y = ax + b$.

(ii) Write the relation and find the bill for 35 GB.

(iii) What do a and b represent?

END OF QUESTION PAPER — Total: 40 Marks